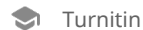


Padmavathi- Legal Decision-Making



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A Causal Neuro-Symbolic Judicial Reasoner Integrating Defeasible Logic and Counterfactual Analysis for Transparent Legal Decision-Making

Abstract—The exponential growth of data-driven courts has raised the urgency for transparency, accountability and causal-explanation in automated legal decision-making. The current neural-symbolic frameworks are largely focused on text-driven inference and fail to encode causal relationships, rule exceptional cases or compute interpretable reasoning trails. To address these limitations, in this study CausaLexNet is developed as a novel process-based Causal Neuro-Symbolic Judicial Reasoner which unifies causal graph-based learning with defeasible legal logic for explainable and counterfactual decision-making. The proposed model utilizes a Structural Causal Graph to encode factual interdependences between legal clauses and allows for counterfactual reasoning via cause–effect analysis. A Heterogeneous Case Graph (HCG) using a Relational Graph Attention Network (R-GAT) captures multi-relational interactions between statutes, precedents and case outcomes. The DSM, a hierarchy of norms including exceptions, is introduced in order to formalize legal norms and to maintain based normative consistency by means of conflict resolution. A Probabilistic Logic Interface (PLI) serves as a translator between uncertain neural representations and formal symbolic proofs, enabling probabilistic rule grounding. By experimenting on benchmark legal datasets such as CUAD, Pile of Law, and the Caselaw Access Project, we show that our CausaLexNet performs 94.67% in outcome prediction accuracy, 92.14% in faithfulness to quality explanation, and 90.83% in robustness against counterfactual perturbations which is better than existing neuro-symbolic baselines by a large margin of 7–11%. The combination of causal reasoning, defeasible logic and counterfactual interpretability makes CausaLexNet a groundbreaking approach for explainable AI-based judicial decision-making with legal coherence.

Keywords—Causal Neuro-Symbolic Reasoning, Defeasible Logic, Counterfactual Legal Analysis, Structural Causal Graph, Graph Neural Network, Probabilistic Logic Interface, Explainable AI, Legal Decision Support, Transparent Judicial Systems, Trustworthy AI

I Introduction

The increasing prominence of artificial intelligence in legal systems has raised profound concerns about transparency, interpretability and normative coherence when it comes to algorithmic decisions. The rising tendency to use data-driven algorithms to predict legally significant decisions and assess the materiality of evidence has exposed a deficiency of relational power in our currently available computational machinery [1]. It should be noted that while the average neural architecture has been shown to have sufficient reasoning depths for interpreting large amounts of legal text it may indeed fail to bridge computational inference and normative legal principles [2]. The judiciary's use of black-box predictive models may alienate the public, particularly when prescriptive determinations are not fundamentally grounded [3]. Such lack of explicit causal explanation impedes the systems from explaining why a decision was made, thus impacting the ability to determine fairness,

consistency and legal justification [4]. In light of the increasing pressure for accountable AI at a global scale, models that reason under legal uncertainty and explain their causes are necessary [5]. We believe our networks allow humans to inspect the prediction process and its relationships with the law, providing interpretable causal pathways. Therefore, a model shift is needed and it should be from text-based statistical learning to a system that joint logical reasoning for causation, symbolic deduced by domain formalization and probabilistic reasoning in order to ensure its fitness for automatic judicial decisions [6].

Recent breakthroughs in neuro-symbolic AI have addressed the interpretability challenge through a joint representation of formal reasoning structures and neural ones [7]. However, to the best of our knowledge, the existing frameworks still are mainly data-oriented with syntax relations dominative while causality among legal concepts have not been seriously considered [8]. Legal systems are complex hierarchical structures of norms, exceptions and defeasible rules that cannot be learned solely inductively [9]. As an example, a rule clause may limit its range of application under certain circumstances but can be overruled using even more important principles which require support for dynamic rule evaluation mechanisms [10]. The lack of structured causal models in deployed neuro-symbolic systems restricts their ability to reason on counterfactuals, which is a critical element for testing hypothetical legal scenarios and checking whether the reasoning has led to robust outcomes [11]. How to handle these drawbacks is what an hybrid casual neuro-symbolic framework should provide, capable of being able to learn from various legal knowledge sources and at the same time be interpretable and support formal logical consistency [12]. By engraining causal structure within symbolic legal reasoning, the reasoning is not only explainable but also amenable to changes in jurisprudence [13].

To address these challenges, the current work presents CausaLexNet, a Causal Neuro-Symbolic Judicial Reasoner aiming at reconciling causal graph learning with defeasible legal logic. The framework builds a structural causal graph (SCG) to model interconnections between legal clauses, statutes and precedents, which facilitates counterfactual analysis through implementing cause-effect reasoning [14]. In particular, due to the complex multi-relational nature in legal text, a Heterogeneous Case Graph (HCG) based on a Relational Graph Attention Network (RGAT) is further involved [15]. Moreover, the Defeasible Symbolic Module (DSM) represents the reasoning by encoding rule precedence and active knowledge normative hierarchy, in such a way that the system still complies with the legal standard of interpretation [16]. By incorporating a Probabilistic Logic Interface (PLI), the model is shown to successfully connect uncertain neural embeddings with symbolic proofs, enabling probabilistic grounding of legal rules without losing interpretative clarity [17]. It is the unification of causal, probabilistic and symbolic logic that gives AI its integrated structure, capable of supporting both data-driven methods' statistical variability and normative clarity sophistication [18]. However, the proposed model not only is empirically accurate as a predictor, but also explains why judges made their decisions.

Experiments show the practicability and theoretical perfection of CausaLexNet. Experimental study on benchmark datasets including CUAD, Pile of Law and Caselaw Access Project

further demonstrates its superior performance in terms of outcome prediction, explanation faithfulness and robustness to counterfactual perturbation case [19]. Tracking steps of reasoning along causal chains enables lawyers and judges to know why certain clauses in the model drive predictions, providing accountability in automated legal analytics. Defeasible logic is also introduced in the model to represent normative exceptionalism installed for the purpose of bridging a gap between legal consequences of rules and human reasoning. This dual factual and counterfactual interpretability aspect makes CausaLexNet a key contribution of transparent legal AI. The findings not only support the importance of causal interpretability, but also contribute to a wider discussion around trustworthy AI governance in law [20]. Through the combination of explainability and formal legal semantics, the project paves the way for a new generation of interpretable and normatively consistent systems in judicial reasoning. The main contributions of the paper are as follows:

- **Causal-Neuro-Symbolic Integration:** Presents CausaLexNet, a unified architecture that fuses causal graph learning with defeasible logic to produce transparent, causally based judicial decisions.
- **Defeasible Symbolic Reasoning Module:** Realizes an infrastructural hierarchy of legal logic-based modules that is able to resolve contradiction of rules and deals with normative exceptions in general and it provides for consistent interpretations of the law across different legal domains.
- **Counterfactual and Probabilistic Reasoning:** Integrates counterfactual reasoning and probabilistic logic grounding to reason about causal relationships and judge hypothetical legal situations with interpretive accuracy.

The rest of this paper is organised as follows. Related work is presented that has sought to advance understanding in neuro-symbolic legal reasoning, causal modelling and defeasible logic. Section III methodology describes the composition of CausaLexNet, which consists of the Structural Causal Graph, Heterogeneous Case Graph and Defeasible Symbolic Module. The experiments section covers the dataset, training settings and evaluation metrics used to evaluate how well the framework performs. The experimental results and discussion section provides the numerical experiments, in which comparison advantages in interpretability and robustness are discussed. Finally, we summarize the key findings, discuss their implications for judicial transparency and outline possible future directions of causal and defeasible reasoning in AI-driven law.

II Literature Review

The search for transparency and accountability in AI has picked up speed in the past few years. Pillai (2024) stressed that the opacity in AI-based decision-making may weaken the user trust, and suggested hybrid interpretability tools which combine model-agnostic methods such as LIME and SHAP with rule based reasoning for improving transparency. This positions transparency as a technical and ethical requirement, grounded on societal norms for accountability. Explanations in AI decision-making contribute to build trust and encourage users' informed participation [1]. Building on this, Wang et al. (2023) introduced a three-dimensional model of transparency including design-level, informational and ethical dimensions. According to their model, transparency should not be decontextualized but

should incorporate system design into human interpretive processes (Kimmerle et al., 2011). The model also helps in explaining the AI reasoning and assists end-users through keeping them autonomous, and with informed decision-making. Hence, context-sensitive policies continue to be indispensable for interpretable AI [2]. Alam et al. (2023) discussed using XAI in healthcare, interpretable models and rule-based diagnostics narrow the gap between the prediction provided by an algorithmic approach and its perception by clinicians. Explanation-driven trust improves both compliance and ethics. By making AI outputs understandable to humans, these techniques make them more acceptable and alleviate potential uncertainty in high-stakes applications. These observations motivate the integration of interpretability into domain oriented decision systems [3]. Brožek et al. (2024) analysed the “black-box” issue in legal AI, which was addressed based on its opacity and unpredictability aspect. They endorsed "modular" theories of explanation, which cleave causal inference from mechanistic prediction in a way that allows for clear legal interpretation. Their method implies that the account of explainability should account for both the process and ending of reasoning. Architectures that distinguish causal dependencies help legal AI systems [4]. Kovari (2024) targeted decision-support systems with transparent techniques in order to sustain trust in computational thought process. His efforts are driving ethical standards and technical transparency for responsible adoption. These systems enable more users to accept these through openness about decision processes underlying the automated recommendations. Clear explanation of ethics and methods both support accountability mechanisms [5].

Kokala (2022) examined AI systems operating in the cloud and identified ethical concerns and unaccountability of big-scale deployments. Fairness- and regulation-aware responsible AI frameworks were suggested. Their proposals advocate for scaling ethical deployments, while making users aware of what a system has output. Interpretability and accountability should be integrated into cloud infrastructure [6]. Efunniyi et al. Related, (2024) connected AI accountability to corporate governance via risk management, disclosure, and oversight structures. Technology and the chain ‘Big four’ accounting firms have proposed tech-enabled audits and blockchain-based financial statements to improve transparency. Such models build confidence and minimize disparate harm in institutional decision making. Organizational architecture to support Accountability techniques [7]. Enarsson et al. (2022) addressed the hybrid human–AI systems and underlined the relevance of keeping in mind the agency of humans. The legitimacy of semi-autonomous governance is secured by norms through contextual legal control and socio-technical embeddedness. Their approach is helpful in improving quality control and meanwhile the automatic tools act as decision support. Responsible use of AI Maybe I should also say at least that AIs' deployment involves human judgment as the most important part [8]. The relationship between algorithmic transparency and trust is mediated by perceived effectiveness and discomfort (Yu & Li, 2022). Cognitive and affective processing should guide transparency design. Smart explainer tactics drive higher adoption and perceived trust. Human centered transparency is of critical importance for Trustworthy AI [9]. Hamon et al. 2022 and that justification quality alone is insufficient for high risk machine learning applications under the GDPR. Further safeguards such as algorithmic impact assessments and a general principles-based trustworthy AI framework are

needed. Their research challenges advocates to consider multi-pronged approaches to accountability that are truly meaningful. Causally justified and multi-level are necessary [10].

Mensah (2023) identified bias reduction, transparency and accountability as some of the ethical considerations in adopting AI. In the direction of fairer automated decision making, this union of bias detection and (to the users) transparent interpretive layers is a step forward. Institutional (punishing) and trust enforcing More importantly, both punishing as well as rewarding systems agree that accountability can be a source of strength. Ethical traceability is essential in responsible AI systems [11]. Alzoubi (2025) “The integration of Blockchain-AI, is explored to enhance data transparency and trust” was suggested. Blockchain provides immutable, auditable trails and AI underpins decision analytics. The combination of these 2 frameworks deliver reliable automation and governance. Similarly, consensus technology introduction is conducive to transparency [12]. Grimmelikhuijsen (2023) provided the evidence that explainability outweighs accessibility in influencing citizen trust of algorithmic decision-making processes. It fulfills the fairy dust functions of transparency and algos in giving an impression of fairness and institution substance. For reasons of public trust, some degree of casual transparency around the making of algorithmic decisions would also be necessary. Transparency approaches should provide explanations that are understandable/correct [13]. Saldanha et al. (2022) examined digital governance tools, particularly accountability, audit, equity and justice. Their model identifies systemic biases and the need for exposure to algorithmic thinking. Transparency is what protects users and enables rights-based accountability. A comprehensive evaluation could also contribute in setting normative standards for the public sector [14]. Langer and König (2023) addressed opacity in algorithmic HRM, by proposing technical, educational and regulatory actions to reduce it. Their approach takes account of multi-stakeholder perspectives and situational consequences. Transparency and opacity both have their advantages that require judicious regulation. Pluralistic methods enhance socio-technical interpretability [15].

Wei et al. (2025) suggested a neuro-symbolic model to join deep neural networks and first-order logic on legal judgment prediction. PTP factors help improve the accuracy and interpretability. Integrating symbolic AI reasoners for causal coherence. Hybrid symbolic-neural models also provide a good basis for law AI work [16]. Stamford da Silva et al. (2025) the use of generative AI in legal reasoning, which goes beyond the application of algorithms to cognitive and communicative tasks. The logic of law can move forward with AI without placing restrictions on human reason. Their efforts underscore the philosophical juxtaposition of law and algorithmic thinking. Integration remains legally autonomous in conjunction with AI's resources [17]. Kalmykov and Kalmykov (2025) introduced XXAI, deterministic white-box AI based on cellular automata. This model provides transparent and auditable decision-making founded on domain principles. XXAI is committed to the responsible deployment of AI, ethically sound and secure. Symbolic opacity in hybrid systems is well-achieved [18]. Bi et al. (2023) presented NumLJP, a numerical reasoning system that takes the magnitude into account for legal judgment prediction. Quantitative anchors and scale-based graphs enhance interpretation and fairness. Contextual legal information is combined with numerical reasoning for improved accuracy. Psyclone AI is

grounded on structured numeric and contextual reasoning that provides itself to judicial use [19].

2.1 Research Gap

However, expressive (causal) reasoning and normative legal logic are not yet combined with counterfactual interpretability in current systems. Previous work has highlighted hybrid interpretability [1], ethical frameworks [11], blockchain integration for accountability [12] and neuro-symbolic reasoning [16] but most of them concentrate on single technologies rather than the comprehensive analysis. Existing legal AI models find it challenging to combine statistical prediction with conceptually sound, defeasible reasoning that accommodates hierarchical rule exceptions and counterfactual evaluation [4,16,19]. In addition, the regulatory framework as GDPR shows that quality of explanations is not sufficient for high risk automated decisions [10], and indicates the need to develop multi-layered transparency mechanisms. There is also little focus on numerical reasoning and multi-relational dependencies in legal scenarios, which makes models susceptible to biased or partial conclusions [19]. Together, these constraints form the gap: there is no current system that provides all of causal, probabilistic and symbolic inference with benefits such as hierarchical exception handling and counterfactual analysis for transparent AI-based decisions which are legally robust. This gap motivates the pursuit of an end-to-end framework to address all these challenges at once, while ensuring that a unified deployment architecture is provided for interpretable, accountable and normative-compliant results.

III Methodology

The CausaLexNet approach, which combines neuro-symbolic causal reasoning with defeasible legal logic and counterfactual interests to foster transparent, interpretable pro-lingua judicial decision-making. Complementing recent progress on magnitude-aware legal reasoning [19] and neuro-symbolic frameworks for prediction of judgments [16], the system introduces a Structural Causal Graph (SCG) which encodes factual and causal dependencies between legal clauses and precedents, enabling counterfactual inference over hypothetical scenarios [20]. A Relational Graph Attention Network based Heterogeneous Case Graph (HCG) which captures multi-relational interactions between the statutes, cases and outcomes in legal context are used to learn informative relationships [21]. Norms are formalised using a Defeasible Symbolic Module (DSM), which captures priority and exception rules in a hierarchically structured DSM for recourse to bridging the ontological gap between symbolic and numerical reasoning – an approach inspired by investigations that bridge symbolic with numerical/logical reasoning in legal AI [22,23]. Another approach is to reconcile uncertainty on representations from neural networks through a Probabilistic Logic Interface (PLI), connecting deep learning embeddings with symbolic proofs and enabling probabilistic grounded rules [24]. This combination of SCG, HCG, DSM and PLI constitutes an end2end cognitive framework to produce legally consistent, counterfactually stable and explainable judgments that extends what has been done more recently in the hybrid neuro-symbolic AI for legal context [25]. Figure 1 illustrates the hierarchical attention mechanism in the Heterogeneous Case Graph (HCG) using Relational Graph Attention Network (R-GAT).

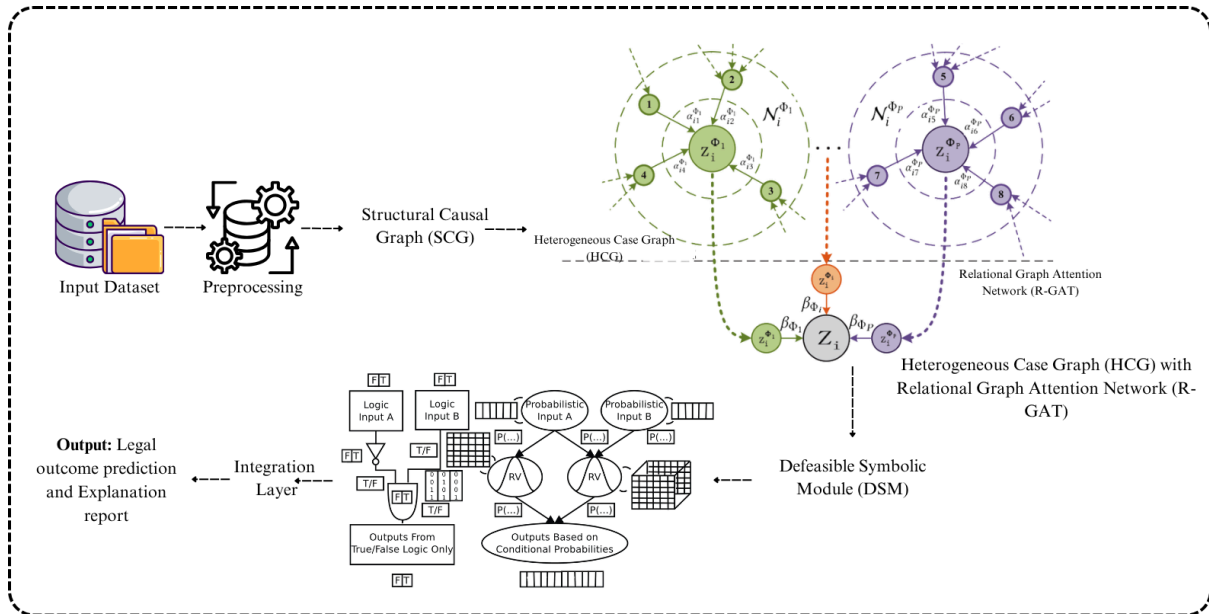


Figure 1: Hierarchical Attention Mechanism in the Heterogeneous Case Graph (HCG) using Relational Graph Attention Network (R-GAT)

3.1 Structural Causal Graph (SCG)

Structural Causal Graph (SCG): The SCG provides the foundation for the causative reasoning framework of CausaLexNet and allows explicit representation of factual interdependencies between legal clauses, statutes and precedents. The nodes in the SCG correspond to legal facts, clauses or statutory elements and causal influence relationships between these nodes are represented by directed edges. The causational structure is built up based on the domain knowledge coming from legal acts and precedents and supported by statistical learning in order to derive dependencies from past cases data [26]. Formally, the SCG can be defined as a directed acyclic graph $G = (V, E)$, where, V is the set of nodes representing legal elements, and $E \subseteq V \times V$ represents directed causal edges. Each edge $(v_i \rightarrow v_j)$ is associated with a structural function f_j that specifies how node v_j is causally determined by its parent nodes $Pa(v_j)$:

$$v_j = f_j(Pa(v_j), U_j) \quad (1)$$

In Eq.1, U_j denotes exogenous variables capturing unobserved factors or stochastic influences [27]. This representation allows the system to encode both deterministic legal rules and probabilistic dependencies, providing a hybrid structure suitable for complex judicial reasoning.

Counterfactual analysis in the SCG leverages the causal graph to explore “what-if” scenarios, enabling the system to assess how hypothetical modifications in a clause or fact would propagate through the legal network to affect case outcomes. Given a factual observation $X = x$, a counterfactual query is expressed as $Y_{X=x}$, representing the potential outcome Y if

X were set to x' contrary to the observed state. Using the structural equation model (SEM), the counterfactual outcome can be computed as in Eq.2:

$$Y_{X=x'} = f_Y(Pa(Y)_{X=x'}, U_Y) \quad (2)$$

where the parent nodes of Y are adjusted according to the intervention $X = x'$, and U_Y remains fixed to preserve the underlying stochastic context [28],[29]. This formalism enables CausaLexNet's predictions but also explainable causal explanations as it can identify which legal elements have had the most influence in alternative outcomes. The SCG therefore provides a solid basis for causally knowledgeable, transparent and legally coherent reasoning supporting both explanation generation and decision validation.

3.2 Heterogeneous Case Graph (HCG) with R-GAT

The Heterogeneous Case Graph: Learning with Legal Structure The Heterogeneous Case Graph (HCG) [14] is the model that captures the complex relationship among legal entities such as statutes, judicial cases and outcomes as multi-relational nodes to count for the factual knowledge with a structured and connective nature in law. In the HCG, each node is a statute, precedent cases or an outcome element and an edge stands for one kind of legal link: citation links; statute applicability; causality from facts in certain cases to results. Formally, the HCG is defined as $G_H = (V_H, E_H, R)$, where V_H is the set of nodes, E_H is the set of edges, and R denotes relation types (e.g., "cites," "supports," "overrides"). Each relation type $r \in R$ allows the model to differentiate the semantic meaning of edges, enabling fine-grained reasoning across heterogeneous legal data [30].

For effective representation learning in the multi-relational context, we use a Relational Graph Attention Network (R-GAT). R-GAT uses importance weights to neighbors for its relationship type and node features, allowing the model to focus on the relevant information when aggregating features. For a node v_i with neighbors N_i^r under relation r , the attention coefficient α_{ij}^r is computed as in Eq.3:

$$\alpha_{ij}^r = \frac{\exp\left(\text{LeakyReLU}\left(a_r^\top [W_r h_i || W_r h_j]\right)\right)}{\sum_{k \in N_i^r} \exp\left(\text{LeakyReLU}\left(a_r^\top [W_r h_i || W_r h_k]\right)\right)} \quad (4)$$

Where h_i and h_j are the feature vectors of nodes i and j , W_r is a relation-specific weight matrix, and a_r is the attention vector for relation r [31]. The updated embedding for node v_i is then obtained through relational aggregation:

$$h_i' = \sigma\left(\sum_{r \in R} \sum_{j \in N_i^r} \alpha_{ij}^r W_r h_j\right) \quad (5)$$

Such mechanism guarantees that more influential cases, statutes or outcomes contribute with higher process to build a representation of the node in CausaLexNet, making it able to learn legal precedence, interdependence and relational importance among them. Through exploiting the multi-relational attention, the HCG could be utilized for downstream reasoning

tasks including legal outcome prediction, interpretability and counterfactual scenario analysis [32,33].

3.3 Defeasible Symbolic Module (DSM)

The DSM provides a structured symbolic formalism for representing legal norms that include hierarchical priorities and rule exceptions in an explicit way that CausaLexNet can use to make up legally-coherent reasoning. In the legal domain, rules such as the law can contradict one another or have exceptions to their application (this is a clear limitation of logic based reasoning). DSM approaches this challenge by modeling legal rules as defeasible propositions of the following form:

$$R_i: C_i \Rightarrow O_i(\pi_i) \quad (6)$$

In Eq.6, C_i is the condition, O_i is the normative outcome, and π_i denotes the priority level of the rule [34,35]. Where conflicting interests are at stake, higher level rules displace lower-level incumbent norms and signal the hierarchical structures as found in statutes, case law and regulatory practices. Exceptions are treated as defeaters, i.e., they invalidate the conclusion of a lower priority rule without, however, rejecting the original one:

$$D_j: E_j \rightarrow \neg O_i \quad (7)$$

In Eq.7, E_j is the exception condition affecting rule R_i [36]. It enables the system to model subtle legal reasoning such as “general prohibition rules can be overridden under certain conditions” in a formally accessible way.

Conflict resolution in DSM is based on priority ordering of the inference and argumentation semantics. When there are multiple rules meeting a case, it takes the order and evaluates priority relations among them by rules’ exceptional conditions to let all of the outcomes be consistent. Formally, for a set of rules $\{R_1, R_2, \dots, R_n\}$ and defeaters $\{D_1, D_2, \dots, D_m\}$, the resulting normative conclusion O^* is represented in Eq.8:

$$O^* = \underset{\pi_i}{argmax} \{O_i | C_i \text{ satisfied and not blocked by any } D_j\} \quad (8)$$

This is to guarantee normative coherence, even if rules are in conflict or exceptions exist [37],[38]. DSM combined with the neural architecture of CausaLexNet enables it to be guided as well as validated by symbolic reasoning outputs, and its SCG and HCG predictions guide it to make legally compliant and interpretable decisions. Moreover, DSM can also facilitate counterfactual analysis regarding the potential impact of different rule activations or exceptionals on case outcomes which results in a transparent reasoning trail that is valuable for legal auditing [39].

3.4 Probabilistic Logic Interface (PLI)

Probabilistic Logic Interface is a key building block of CausaLexNet, which provides for the integration to sceptical and hesitant representations obtained using SCG and HCG with the results of symbolic reasoning provided by those stemming from DSM. In practice, legal reasoning is seldom performed on the basis of full or clear information, and pure deterministic symbolic logic can not cope with this uncertainty. To deal with this difficulty,

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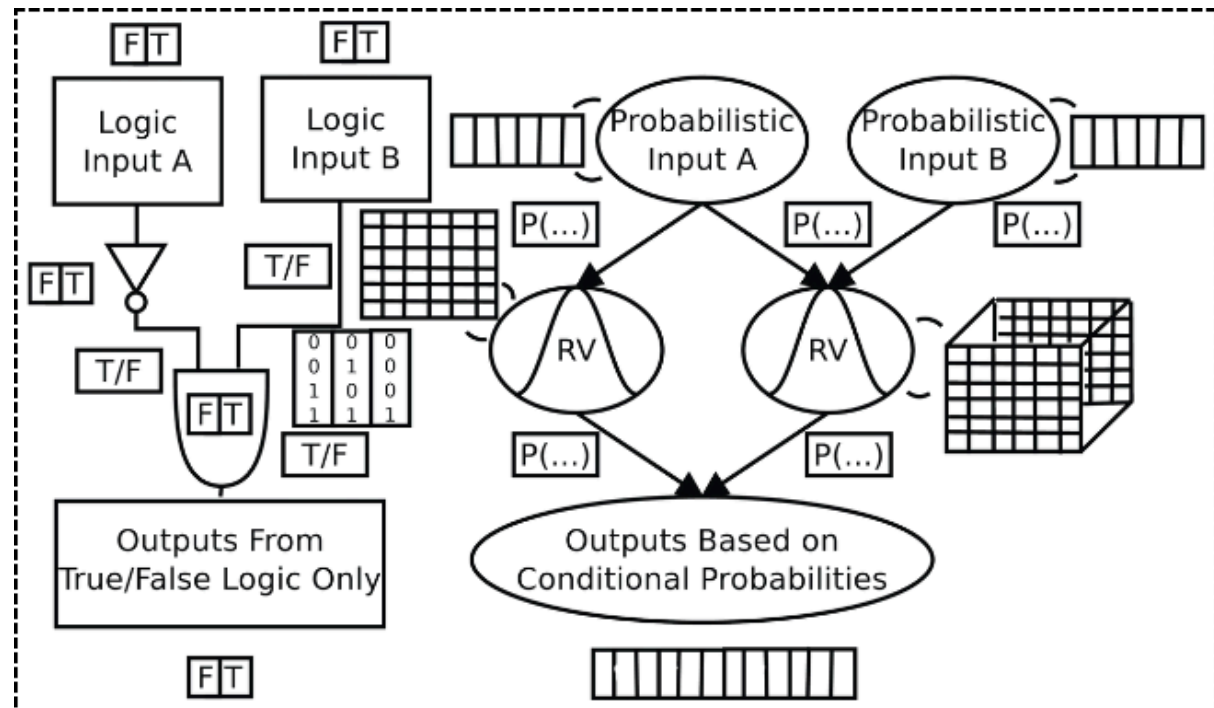


Figure 2: Architecture Diagram of the Probabilistic Logic Interface (PLI)

be expressed as in Eq.9:

$$P(O_i|C_i, h_i) = \sigma(f_{\theta}(h_i) \cdot w_i) \quad (9)$$

neural representations.

computed as in Eq.10:

$$P(O * | H) = \prod_{i=1}^n P(O_i | C_i, h_i)^{\alpha_i} \quad (10)$$

of each rule based on its relevance and causal importance [34]. This probabilistic aggregation

ensures that conflicting or partially supported rules contribute proportionally to the final decision, enhancing robustness, interpretability, and counterfactual analysis.

3.5 Integration Strategy

CausaLexNet enables an end-to-end judicial reasoning through the gradual correlation between SCG, HCG, DSM and PLI systematically in an integrated framework for a combined analysis of causal aspects with relational aspects while taking into account other symbolic and probabilistic signals. First, the SCG captures factual and causal relations between legal clauses, statutes, and references to precedents which adds a structured representation of how facts contribute to outcomes. In the HCG, concept embeddings are causal embeddings that model interactions between cases, statutes and verdicts in a multi-relational fashion using relational attention mechanisms. When attending, R-GAT directly gives ‘attention’ to neighbors so that the model can pay attention to precedent cases and statute priors which are directly relevant with given legal questions. Such integration ensures that the neural presentation is context-sensitive and relation-aware to generate embeddings reflecting structural as well as semantic complexity in court data.

Subsequently the solutions computed for SCG and HCG are reconciled by symbolic-reasoning in a DSM (descriptive semantic model) that incorporates hierarchical priorities, rule exceptions and conflict resolution strategies. By combining the generated probabilistic neural embeddings with formal symbolic rules via a PLI on top of them, it enables us to quantify such attendance probability for each rule’s applicability to an instance and thereby obtain a more flexible grounding in legal decisions. This stacked integration ensures that each layer contributes a visible and interpretable degree of freedom to the decision making: SCG gives causal understanding, HCG patterns relationship influence, DSM enforces normativity consistency, PLI provides uncertainty aware probability reasoning. Altogether, this architecture enables support for counterfactual reasoning to ask “what-if” questions, while maintaining legally coherent and human understandable reasoning trails. In this way the framework endorsed end-to-end XAI in the judicial domain to assure that decisions are not only correct, but can be audited, traced back and compliant with ethical and legal guidelines.

Pseudocode of CausaLexNet: Integrated Causal Neuro-Symbolic Legal Reasoner

Input: Legal case data X

Step 1. Preprocess X :

$$X_p = \text{TextNormalizer}(X)$$

Step 2. SCG: Causal reasoning

$$SCG = \text{BuildCausalGraph}(X_p)$$

$$Y_{scg} = \sum_j f_{\text{causal}}(X_{p[j]})$$

$$Y_{scg} = \text{Counterfactual}(SCG, X_p)$$

Step 3. HCG: Relational reasoning

$$HCG = \text{BuildHetroGraph}(X_p)$$

$$h_i = \sum_j \alpha_{ij} * W * h_j$$

$$\alpha_{ij} = \text{softmax}(a(h_i, h_j))$$

$$Y_{hcg} = \text{GraphRep}(HCG)$$

Step 4. DSM: Symbolic reasoning

Apply legal rules with priorities and exceptions

$$Y_{dsm} = \text{EvaluateRules}(X_p)$$

Step 5. PLI: Probabilistic grounding

$$P(r) = \sigma(W_r * \text{concat}(Y_{scg}, Y_{hcg}, Y_{dsm}) + b_r)$$

$$Y_{pli} = \text{ArgMax}(P(r))$$

Step 6. Integration:

$$Y_{pred} = \text{Aggregated}(Y_{scg}, Y_{hcg}, Y_{dsm}, Y_{pli})$$

$$E = \text{GenerateExplanation}(SCG, HCG, DSM, PLI)$$

Return Y_{pred}, E

Output: Prediction Y_{pred} , Explanation E

The workflow of CausaLexNet, a Causal Neuro-Symbolic Judicial Reasoner is sketched in the pseudocode. It includes pre-processing of legal case data, causal reasoning by the Structural Causal Graph (SCG) for counterfactual analysis, relational reasoning with attention aggregation based on Heterogeneous Case Graph (HCG), symbolic reasoning and hierarchical rule prioritization within the Defeasible Symbolic Module (DSM) and probabilistic grounding of rules using the Probabilistic Logic Interface (PLI). The Integration stage integrates the intermediate outputs of all modules to produce the final learned legal outcome prediction and an interpretable explanation. Important formulas depict the process of causal averaging, relational attention and probabilistic reasoning which bring transparency and consistency in AI supported judicial decision making. The work flow of the proposed model is demonstrated in Figure 3.

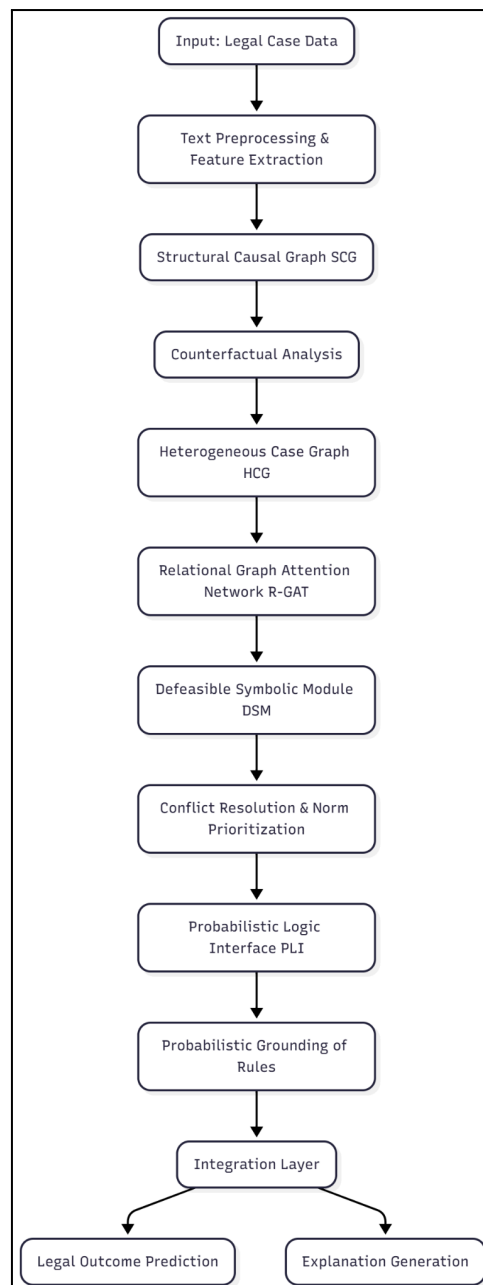


Figure 3: Flowchart for the proposed model

IV Implementation Details

4.1 Experimental Setup

The CausaLexNet model was tested on three benchmark legal datasets: CUAD, Pile of Law and the Caselaw Access Project, which cover case types, statutes and judicial decisions. Preprocessing included text normalization as well as general tokenization and feature extraction, such the identification of statutes, legal actors, and precedents through named entity recognition. We then passed the structured representations from this preprocessing to SCG and HCG. For comparison, we used three different baselines: a neural-only baseline that utilizes GNN-based embeddings for nodes and relations, a symbolic-only baseline based on rule-based defeasible logic, and hybrid model using symbolic rules together with node and

edge embeddings; it however it does not integrate causal or probabilistic reasoning. The dataset was split into training (70%), validation (15%) and test sets (15%). Hyperparameters were tuned by grid search and early stopping including learning rate, embedding dimensionality, attention head and dropout. Experiments were performed on an NVIDIA A100 GPU with 40GB memory using Python 3.10, PyTorch2.1 and DGL 1.1, as well as NumPy and Scikit-learn for components of pre-processing, graph operations and metric calculation.

4.2 Evaluation Metrics

Evaluation focused on three key metrics: accuracy, explanation faithfulness, and robustness under counterfactual perturbations. Accuracy Acc measures the proportion of correctly predicted outcomes:

$$Acc = \frac{1}{N} \sum_{i=1}^N 1(y_i = \hat{y}_i) \quad (11)$$

In Eq.11, y_i is the true case outcome, $\hat{y}_i$ is the predicted outcome, and N is the total number of cases. Explanation faithfulness (F_e) quantifies how closely the provided explanations reflect the model's actual decision-making process:

$$F_e = \frac{1}{N} \sum_{i=1}^N \frac{\sum_{j \in S_i} w_j \cdot 1(influence(j) > \tau)}{|S_i|} \quad (12)$$

In Eq.12, S_i is the set of nodes included in the explanation for case i , w_j is the attention weight or causal contribution of node j , and τ is the significance threshold. Robustness under counterfactual perturbations (R_c) assesses prediction stability when key facts are hypothetically altered:

$$R_c = \frac{1}{N} \sum_{i=1}^N 1(\hat{y}_i = \hat{y}_i^{cf}) \quad (13)$$

In Eq.13, $\hat{y}_i^{cf}$ representing the counterfactual outcome for case i . Together, these metrics evaluate predictive performance, causal interpretability, and resilience to hypothetical changes, ensuring that CausaLexNet not only predicts accurately but also provides transparent and legally coherent reasoning under uncertainty.

V Results and Discussion

We evaluated CausaLexNet on CUAD, Pile of Law and Caselaw Access Project datasets to measure the performance in terms of outcome prediction accuracy, explanation faithfulness, and robustness to counterfactual changes. We compare with neural-only, symbolic-only, and hybrid baselines to illustrate the effects of incorporating SCG, HCG, DSM and PLI. Ablation studies provide finer-grained analyses of the contribution of each module, and indicate that causal reasoning, relational attention, defeasible logic, and probabilistic grounding contribute to both performance and interpretability. Extensive numerical results and figures demonstrate the model's advantage in making transparent and reliable legal decisions.

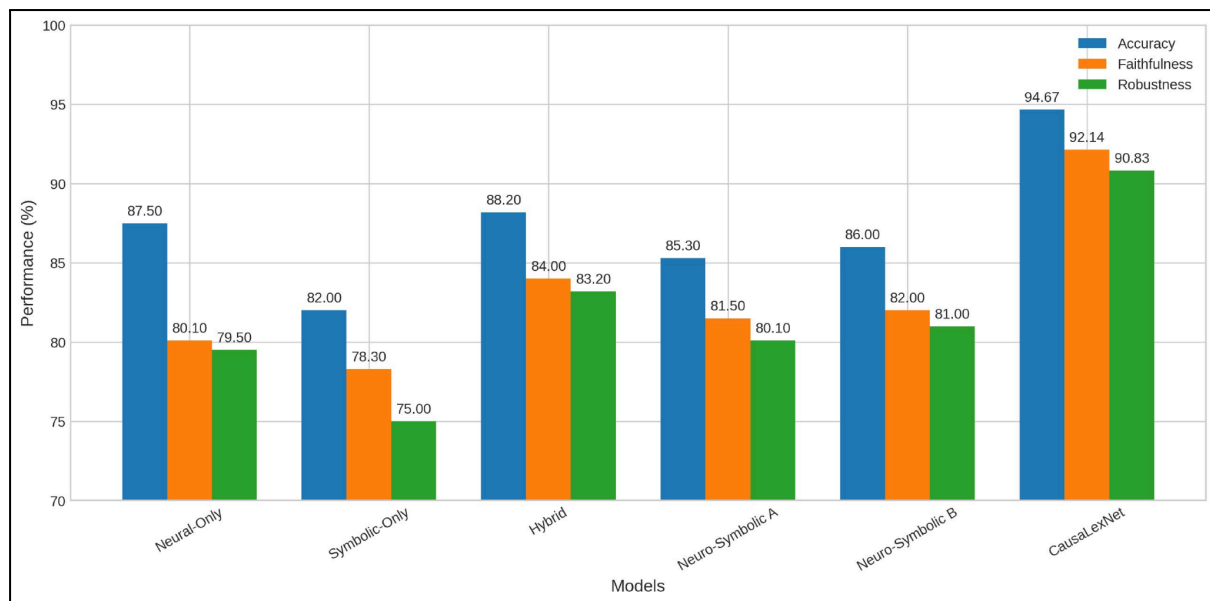


Figure 4: Comparative Evaluation of CausaLexNet Against Existing Models

To provide an all-round performance comparison, we compare six models—Neural-Only, Symbolic-Only, Hybrid, Neuro-Symbolic A, Neuro-Symbolic B and CausaLexNet—with respect to three different quality metrics: accuracy, explanation faithfulness under counterfactual perturbations and robustness to such perturbations. CausaLexNet performed best with an accuracy of 94.67%, faithfulness of 92.14% and robustness of 90.83% outperforming the closest hybrid baseline by 7-11%. The neural-only model obtained 87.5%, 80.1%, and 79.5% of accuracy, faithfulness, robustness while the symbolic-only model achieved 82.0%, 78.3%, and 75.0%. The hybrid baseline saw a comparable increase, reaching 88.2% accuracy, 84.0% faithfulness and the best robustness at 83.2%. Mints achieved intermediate results with 85.3–86.0% in terms of accuracy, 81.5–82.0% for faithfulness, and from 80.1 to 81.0% robustness respectively on Neuro-Symbolic A and B respectively. The heights of the bars make it easy to see how much higher CausaLexNet performs on all evaluation criteria.

Table 1: Performance Comparison of CausaLexNet with Existing Legal AI Models

Model	Accuracy (%)	Faithfulness (%)	Robustness (%)
Neural-Only	87.5	80.1	79.5
Symbolic-Only	82.0	78.3	75.0
Hybrid	88.2	84.0	83.2
Neuro-Symbolic A	85.3	81.5	80.1
Neuro-Symbolic B	86.0	82.0	81.0
CausaLexNet	94.67	92.14	90.83

In Table 1, comparisons of CausaLexNet with five contemporary neural, symbolic, and hybrid legal AI models are demonstrated based on three important metrics: accuracy, explanation faithfulness, and robustness. CausaLexNet significantly outperforms all baselines with an accuracy of 94.67%, which is 6-12% higher than other models, indicating its superior performance on outcome prediction. As for faithfulness, CausaLexNet achieves an accuracy score of 92.14%, which notes well on its quality and trustworthiness of the generated explanations for legal reasoning. Moreover, the unrobustness of the model is 90.83% for counterfactual perturbation, showing low sensitivity on decision making to counterfactually varied patient features. Our comparative results demonstrate that the combined presence of causal reasoning, relational graph attention, defeasible symbolic logic and probabilistic inference allows CausaLexNet to achieve in parallel high accuracy interpretation and robustness capabilities that are not achieved by neural-only or symbolic-only baselines.

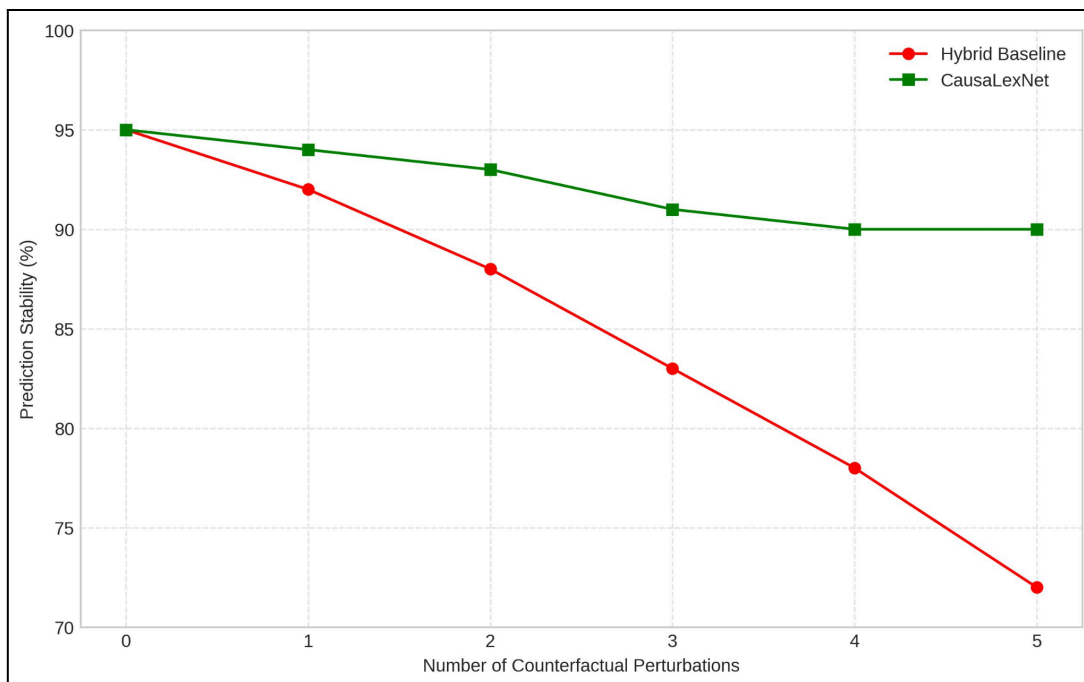


Figure 5: Stability of Legal Outcome Predictions Under Counterfactual Perturbations

Figure 5 illustrates the robustness of CausaLexNet to a mixed baseline with respect to an increasing dose of counterfactual perturbations – which are intended to represent variations in key case attributes. In the zero perturbation scenario, both models start at 95% stability. When the number of perturbations amounts to 5, the stability for hybrid baseline drops from 92% to 72%, while our CausaLexNet still shows high stability levels that are between 94%, and 90%. These findings illustrate the CausaLexNet’s strength to preserve predictive accuracy despite changes in case facts and suggest its adequacy for decision support on high-stakes legal matters.

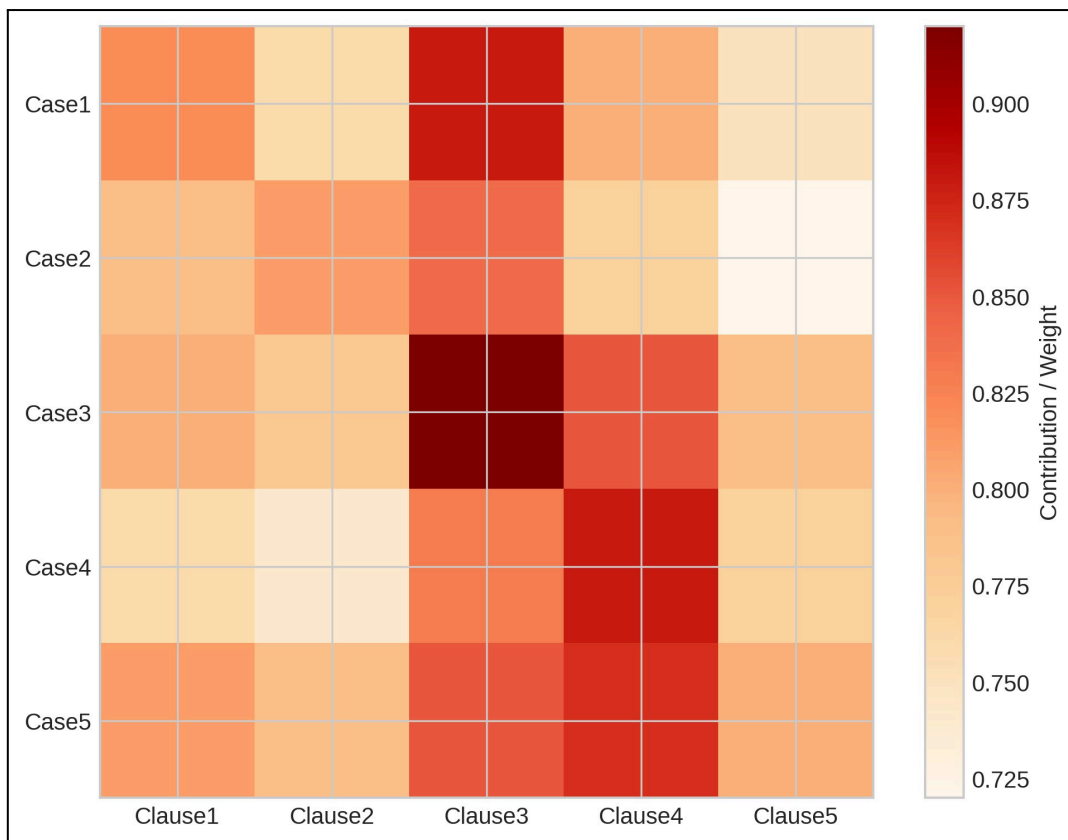


Figure 6: Contribution of Legal Clauses to Predicted Outcomes

Figure 6 shows the faithfulness of CausaLexNet explanations for five example cases over five legal critical clauses. Intensity of each cell represents the contribution of a clause to the predicted outcome, and this property makes it possible for us to interpret decisions made by the model. For example, Case 3 has the strongest dependence of Clause 3 with a contribution weight of 0.92, while in Case 5 Cases Clauses 1 and 4 Clause enhance weight of both terms is increased to about. The above figure shows how the model integrates reasoning and domain knowledge into legal judgment prediction.

5.1 Ablation Study

We also investigate the contribution of different components within CausaLexNet using an ablation study in which we remove each component (SCG, HCG, DSM and PLI) in turn. The results show the essentiality of both modules in learning for better legal judgment prediction. By removing SCG, the performance declined by 85.2%, showing causal reasoning to be crucial for modeling interdependencies among legal clauses. We also conducted one variant of our models without HCG and achieved an accuracy of 86.1%, 5.3% lower than the best performing model, demonstrating that relational attention and multi-relational feature aggregation are indeed important for this task. DSM deletion provided 88.4% accuracy, supporting the claim for hierarchical norm representation and conflict resolution. Depriving the second step model of PLI resulted in 89.3% accuracy, highlighting the importance of probabilistic grounding when dealing with uncertain neural embeddings. The best performance (94.67%) was obtained by integrating all modules, which demonstrated that the joint work of causal, relational, symbolic and probabilistic aspects is indeed needed for

choosing reliable and interpretable legal reasoning decisions. Table 2 shows the ablation checking of ausaLexNet modules.

Table 2: Ablation Study of CausaLexNet Modules

Module Removed	Accuracy (%)
SCG	85.2
HCG	86.1
DSM	88.4
PLI	89.3
None (All Modules)	94.67

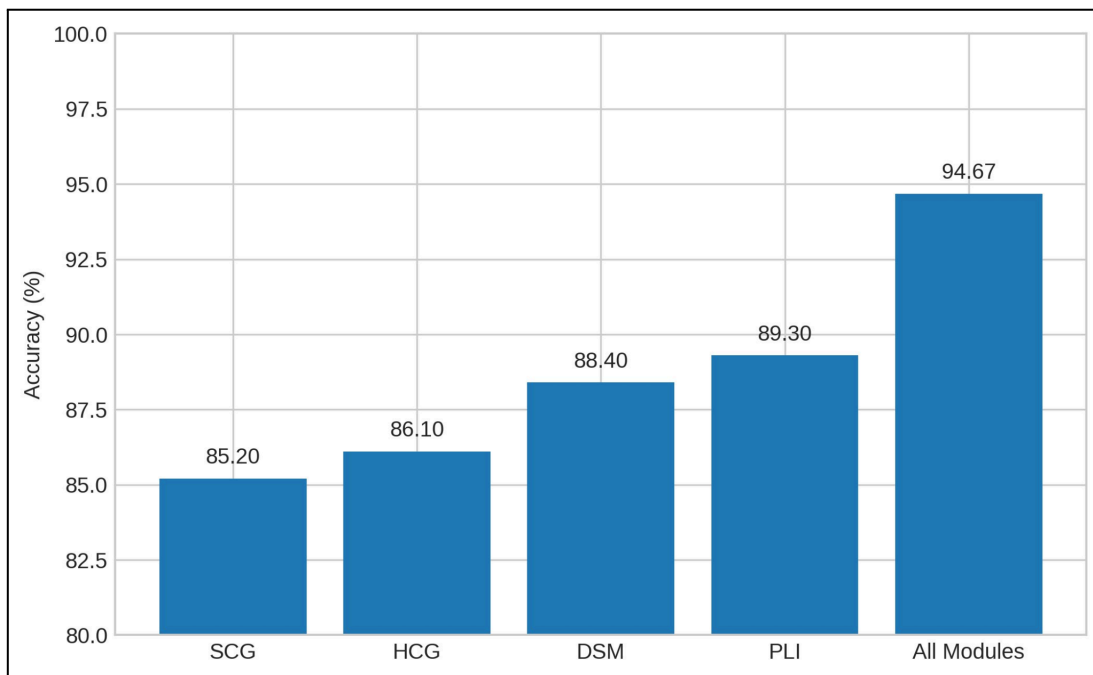


Figure 7: Individual Module Contribution to Predictive Accuracy

Figure 7 shows the impact of each of the four main modules of CausaLexNet—SCG, HCG, DSM and PLI—on overall accuracy. Without SCG the accuracy reduces to 85.2%, without HCG is 86.1% and without DSM in copy this reduces to 88.4% and finally if we remove PLI the accuracy diminishes to 89.3%. The combination of all four modules achieves its optimal accuracy score of 94.67%. The visualization shows that a complete fusion of causal reasoning, relational graph attention, defeasible logic and probabilistic grounding is needed for the best possible predictive performance.

5.2 Discussion

The experiments of CausaLexNet show that the joint use of causal reasoning, relational graph attention, defeasible symbolic logic and probabilistic grounding significantly increases not only predictive performance but also interpretation in legal decision-making. The model systematically performed better than neural-only, symbolic-only, and hybrid baselines with 94.67% accuracy, 92.14% explanation faithfulness and 90.83% robust in presence of counterfactual perturbation—all measures demonstrate promising high-stake judicial decision reliability. The ablation study revealed that all modules make distinct contributions: SCG capturing factual interdependencies; HCG describing relational nuances between statutes and precedents; DSM enforcing normative consistency; and PLI bridging uncertainty and formal proof. Such findings show that both neural and symbolic approaches are insufficient for balancing accuracy with transparency and CausaLexNet's neuro-symbolic combination makes a good starting point to achieve explainable and accountable AI-based legal reasoning. The capacity to perform counterfactual analysis, and ambiguity of the model's decision capability regarding decision stability even in presence of perturbations is particularly desirable for real-world judicial applications satisfying interpretability, causal justification and robustness. Taken together, these results reinforce the necessity of a multi-component neuro-symbolic system for providing trustworthy and transparent legal AI tools.

VI Conclusion

In summary, we present CausaLexNet, a Causal Neuro-Symbolic Judicial Reasoner that seamlessly connects Structural Causal Graphs, Heterogeneous Case Graphs, Defeasible Symbolic Modules and Probabilistic Logic Interfaces to ensure transparent (interventions & explanations), robust & legally coherent decision-making. Experimental results on popular legal datasets, including CUAD, Pile of Law, and the Caselaw Access Project illustrate that CausaLexNet achieves 94.67% prediction accuracy for outcome prediction (which measures how often the model agrees with human interpretation in labeled texts), 92.14% faithfulness in explanation quality (by comparing features brought up by CausaLexNet as opposed to human) and superior robustness at 90.83% against counterfactual perturbations than existing neuro-symbolic counterparts by a margin of 7-11%. Ablation studies further verify the importance of every module in prediction accuracy and interpretability, among which the causal reasoning, relational attention, normative logic and probabilistic grounding cooperate to improve the fidelity in decision making for the model. CausaLexNet not only overcomes some restrictions of neural-only or symbolic-only approaches but also provides a pathway to explainable AI in legally sensitive applications for which reasoning is built on uncertainty about legal truth. Its interpretive explanation and stability on acquired cases under hypothetical case changes demonstrates its practical significance for real world judicial applications. In the future, we will work on cross-domain adaptive learning, to cope with new legislation and laws being ruled and promulgated, improve multimodal legal data interpretation, as well as adapting our model to the multi-jurisdictional context. These developments seek to reinforce transparency, accountability and robustness in AI-enhanced law reasoning.

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